

HATCHER–THURSTON COMPLEX FOR SURFACES WITH NON-PLANAR ENDS

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ABSTRACT

In this paper, for each $k \in \mathbb{N}$, we define a complex $\Gamma_k(S)$ for an infinite-type surface S with non-planar ends, which serves as an analog of the Hatcher–Thurston complex for the infinite-type setting. We show that $\Gamma_k(S)$ is connected, simply connected, and that the automorphism group of $\Gamma_k(S)$ is isomorphic to the extended mapping class group.

INTRODUCTION

All surfaces in this paper will be connected, orientable, without any punctures or marked points, and possibly with non-empty compact boundary. The mapping class group, $\text{Map}(S)$, is the group of all orientation preserving homeomorphisms of the surface S up to isotopy. The extended mapping class group, $\text{Map}^*(S)$ is the group of all homeomorphisms of S up to isotopy. When the fundamental group $\pi_1(S)$ of the surface S is finitely generated, we say that S is of finite-type; otherwise S is infinite-type.

Recently, there has been an effort in the literature to generalize results about curve graphs and complexes from the finite-type case to the infinite-type case. In [BDR20], the authors show that the automorphism group of the curve complex, $\mathcal{C}(S)$, of an infinite-type surface with empty boundary is geometric, that is, every automorphism of $\mathcal{C}(S)$ is induced by the action of $\text{Map}^*(S)$, generalizing the result of [Iva97, Luo00, Kor99] for finite-type surfaces. Independently, [HHV17] also proved that the automorphism groups of the curve complex, non-separating curve complex, and the Schmutz graph are all isomorphic to each other and to $\text{Map}^*(S)$ for surfaces of infinite-type with no planar ends and finitely many boundary components. Moreover, [HHV17, BDR20] show that these complexes can differentiate infinite-type surfaces.

In this paper, we generalize the Hatcher–Thurston complex to the setting of infinite-type surfaces with possible compact boundary and only non-planar ends. The Hatcher–Thurston complex or Cut-system complex, $\mathcal{HT}(S)$, is a 2-dimensional CW-complex associated to surfaces of finite-type in which the vertices are maximal cut-systems of the surface, edges correspond to an elementary move between cut-systems, and 2-cells are attached to triangles, rectangles, and pentagons as described in Section 1. This cut-system complex was first introduced in [HT80] to obtain a presentation for the mapping class group of finite-type surfaces. In [HT80], the authors use a Morse-theoretic proof to show that the complex is connected and simply connected. Later, Wajnryb gave a combinatorial proof of this in [Waj99]. In [IK07], the automorphisms of the Hatcher–Thurston complex were shown to be geometric. In this paper, the complex $\mathcal{HT}(S)$ is generalized to a countable collection of complexes $(\Gamma_k(S))_{k \in \mathbb{N}}$ whose vertices are given by cut-systems with exactly k non-separating curves, as opposed to maximal cut-systems, with 1–cells and 2–cells defined in a similar fashion to $\mathcal{HT}(S)$. In case that S is a finite-type surface of genus g , the collection $\{\Gamma_k(S)\}$ is finite with $1 \leq k \leq g$ and $\Gamma_g(S) = \mathcal{HT}(S)$. Precise definitions of these complexes are given in Section 1. In Section 2, we show the following result.

Theorem 1. *Let S be a surface of infinite-type with non-planar ends. Then $\Gamma_k(S)$ is connected and simply connected for each $k \in \mathbb{N}$.*

The proof of this result closely follows the combinatorial proof of [Waj99] with alterations made when required for the infinite-type case. The proof proceeds by induction on k . We first show that the 1-skeleton, $\Gamma_k^1(S)$, is connected and that its diameter is bounded between k and $8k - 4$ using induction on k . To further prove that $\Gamma_k(S)$ is simply connected, we introduce the notions of radius of a path in $\Gamma_k(S)$ and segments, following the proof by Wajnryb. The remaining proof can then be divided into three steps. It is first shown in Proposition 2.6 that $\Gamma_1(S)$ is simply-connected by first showing that closed paths with radius 1 are contractible. Note that this argument uses the assumption that S has at least one non-planar end. Continuing the induction on k , the second step shows that any closed 0-radius path is contractible in $\Gamma_k(S)$, see Proposition 2.9, by induction on the number of segments in the path. Finally, again using the result about radius 0 paths and the assumption of a non-planar end the theorem statement follows.

Remark. At multiple points throughout the proof of Theorem 1 the existence of at least one non-planar end is invoked, leading one to ask whether this is a necessary condition. In other words, is $\Gamma_k(S)$ simply connected for all $1 \leq k \leq g$ when S is a genus g surface? Since $\Gamma_g(S) = \mathcal{HT}(S)$ for finite-genus surfaces, as mentioned earlier, it follows that $\Gamma_g(S)$ is simply-connected. But the proof of this, in [Waj99, HT80], makes use of the maximality of the cut-systems with g curves in a genus g surface. To the author's knowledge, this question does not seem to be answered in the literature.

In Section 3, we show that automorphisms of $\Gamma_k(S)$ are geometric:

Theorem 2. *Let S be a surface of infinite-type with no planar ends and empty boundary. Then $\text{Aut}(\Gamma_k(S))$ is isomorphic to $\text{Map}^*(S)$ for each $k \in \mathbb{N}$.*

We prove this by constructing an action of $\text{Aut}(\Gamma_k(S))$ on the Schmutz graph, $\mathcal{G}(S)$, as done by [IK07]. The proof in [IK07] is almost entirely transferable to the infinite-type case, but parts of it are reproduced here for the sake of completeness. As a direct corollary to Theorem 2 we have the following:

Corollary 3. *Let S, S' be surfaces of infinite-type with no planar ends and empty boundary. Let $\varphi : \Gamma_k(S) \rightarrow \Gamma_k(S')$ be an isomorphism. Then S is homeomorphic to S' .*

The Hatcher–Thurston complex is an important tool in the study of mapping class group of finite-type surfaces. By studying the stabilizers of vertices, edges, cells in $\mathcal{HT}(S)$ of the action of $\text{Map}(S)$ Wajnryb gave a simple presentation of $\text{Map}(S)$ in [Waj83], which was inspired by a similar approach to the problem by Harer in [Har83]. Harer in the same paper uses a simpler version of the Hatcher–Thurston complex to compute the second cohomology of the mapping class group. These methods are not directly transferable to Big mapping class group as they are not finitely generated, in fact they are not even countably generated. In [PV18], the authors give topological generators for the pure mapping class group $\text{PMap}(S)$, the subgroup of $\text{Map}(S)$ which are end-preserving, in terms of Dehn Twists and handle shift maps. For infinite-type surfaces with countably many non-planar ends it may be possible to use the results presented in this paper to determine relations in $\text{PMap}(S)$ between Dehn Twists and handle shift maps by applying the techniques of Wajnryb and Harer.

Outline. In Section 1 the main objects of this paper are defined and conventions and notations which used throughout the paper are discussed. Section 2 is dedicated to the proof of Theorem 1

and Section 3 to the proof of Theorem 2 and Corollary 3.

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1 PRELIMINARIES

Ends of a Surface. Let S be a surface of infinite-type and let $\{K_i\}_{i \in \mathbb{N}}$ be a strictly increasing chain of compact sets ordered by inclusion which covers S . Then an *end*, e , of the surface S is a strictly decreasing chain $\{U_i\}_{i \in \mathbb{N}}$, where each U_i is a connected component of $S - K_i$. Let $\text{End}(S)$ denote the set of all ends of S . Given an open set $U \subset S$ with compact boundary define:

$$U^* = \{e = \{U_i\}_{i \in \mathbb{N}} \in \text{End}(S) \mid U_i \subset U \text{ for large enough } i \in \mathbb{N}\}$$

Endow $\text{End}(S)$ with the topology generated by $\{U^* \mid U \text{ open subset of } S \text{ with compact boundary}\}$. For any surface, S , the space of ends is homeomorphic to a closed subset of the Cantor set, [AS60]. An end, $e = \{U_i\}_{i \in \mathbb{N}}$, is said to be *planar* if for some $i \in \mathbb{N}$, U_i can be embedded in the plane. Otherwise, the end is said to be *non-planar* or *accumulated by genus*. Denote the subspace of non-planar ends to be $\text{End}_\infty(S)$. The classification theorem for infinite-type surfaces states that S, S' are homeomorphic if they have the same number of genus and boundary components and there is a homeomorphism $\text{End}(S) \rightarrow \text{End}(S')$ which restricts to a homeomorphism of the subspace of non-planar ends, $\text{End}_\infty(S) \rightarrow \text{End}_\infty(S')$, [Ric63]. Throughout this paper, the surface S assumed to have at least one non-planar end.

Curves and Arcs. By a curve α in the surface S we mean a simple, closed curve, that is, an embedding of S^1 in S , which is not null-homotopic and is not homotopic to a boundary component of S . We often consider isotopy classes of curves, which we will denote by the corresponding latin symbol, i.e. $a = [\alpha]$, $b = [\beta]$, $c = [\gamma]$, etc. The *geometric intersection number*, $i(a, b)$, of a pair of isotopy classes of curves a, b is defined as the infimum of the set $\{|\alpha \cap \beta| \mid \alpha \in a, \beta \in b\}$. Curves α and β are said to be in *minimal position* if $|\alpha \cap \beta| = i(a, b)$. Just like in the finite-type case, the curves α, β are in minimal position if and only if they do not form any bigons, [FM11]. Given two isotopy classes of curves a, b there are always representatives α, β , respectively, which are in minimal position. We often abuse notation and call both α and a as curves. The isotopy class, a , of a curve, α , is said to be *separating* if $S - \alpha$ is not connected, otherwise α is said to be *non-separating*. Note that if $i(a, b) = 1$ then both a, b must be non-separating.

Let I be one of $[0, 1], [0, \infty), (-\infty, \infty)$. An arc τ on S is an embedding of $\tau : I \rightarrow S$ so that $\tau(\partial I) \subset \partial S$. An arc is said to be *essential* if it is not isotopic to a boundary or point in S , otherwise it is said to be *non-essential*. Just like curves, we think of arcs in terms of their isotopy classes. The definitions of separating, non-separating, etc can be extended to arcs.

Mapping class group. Let $\text{Homeo}(S)$ be the group of self-homeomorphisms of any surface S then the extended mapping class group $\text{Map}^*(S)$ is defined as $\pi_0(\text{Homeo}(S))$, and elements of $\text{Map}^*(S)$ are called mapping classes and are denoted $[F]$ where $F \in \text{Homeo}^+(S)$ is a representative homeomorphism. Endowing $\text{Homeo}(S)$ with the compact-open topology induces a topology on $\text{Map}^*(S)$. When S is finite-type this is just the trivial topology. This topology is non-trivial when S is of infinite-type. Given a mapping class $[F]$ it induces a map F_* on the collection of isotopy classes of curves: $F_*(a) = F(a)$. It is easily checked that this is well defined. Dehn twist T_a about

a curve a is a mapping class which is represented by a homeomorphism with support inside an annular neighbourhood A containing α given by $(\theta, t) \mapsto (\theta + 2\pi t, t)$.

Curve Complexes and Graphs. For any surface S we define the curve complex $\mathcal{C}(S)$ as the simplicial complex with vertices being the collection of isotopy classes of curves and the vertices $a_1, \dots, a_n$ in $\mathcal{C}(S)$ form an n -simplex if $i(a_i, a_j) = 0$. The non-separating curve complex, $\mathcal{N}(S)$, is the subcomplex of $\mathcal{C}(S)$ with vertex set consisting of only of non-separating curves. The Schmutz graph, $\mathcal{G}(S)$, is the graph with the same vertex set as $\mathcal{N}(S)$, but two vertices have an edge between them if $i(a, b) = 1$. It is a well-known fact that $\mathcal{C}(S)$, $\mathcal{N}(S)$, and $\mathcal{G}(S)$ are connected for any surface, S . In the case when S is of infinite-type all of these complexes have finite diameter, [HHV17]. The action of $\text{Map}^*(S)$ on curves naturally induces an action on these curve complexes via automorphisms.

Theorem 1.1 (Theorem 1, [HHV17]). *Let S be an infinite-type surface with possible compact boundary and no planar ends, that is, $\text{End}(S) = \text{End}_\infty(S)$. Then the action of $\text{Map}^*(S)$ on Γ induces an isomorphism to $\text{Aut}(\Gamma)$, where Γ is one of $\mathcal{C}(S)$, $\mathcal{N}(S)$, $\mathcal{G}(S)$.*

Furthermore, it was shown in the same paper that $\text{Aut}(\mathcal{C}(S))$ is rigid, that is, every isomorphism $\text{Aut}(\mathcal{C}(S)) \rightarrow \text{Aut}(\mathcal{C}(S'))$ is induced by a homeomorphism $S \rightarrow S'$.

Cut Systems and Hatcher–Thurston Complex. A cut system on a surface S is a disjoint collection of isotopy classes of non-separating curves, $(a_0, \dots, a_k)$, so that $S - \bigcup \alpha_i$ is connected. Now, we recall the definition of Hatcher–Thurston complex on finite-type surfaces. Let Σ_g be a finite-type surface with genus g with possible compact boundary. Let $\mathcal{HT}^0(\Sigma_g)$ be the collection of maximal elements of the poset of cut systems on Σ_g ordered by inclusion. Every cut-system in $\mathcal{HT}^0(\Sigma_g)$ has exactly g curves. Cut systems $v, w \in \mathcal{HT}^0(\Sigma_g)$ are said to be related by an elementary move if $v - w = \{a\}$, $w - v = \{b\}$, and $i(a, b) = 1$, we denote this as $v \leftrightarrow w$. Attach a 1-cell between two points $v, w \in \mathcal{HT}^0(\Sigma_g)$ whenever they are related by an elementary move to obtain the graph, $\mathcal{HT}^1(\Sigma_g)$. There are three types of simple cycles we consider:

1. **Triangles:** Consider the vertices $v = (b_0, a_1, \dots, a_{g-1})$, $w = (b_1, a_1, \dots, a_{g-1})$, and $u = (b_2, a_1, \dots, a_{g-1})$ so that $i(b_i, b_j) = 1$. Then the cycle, $v \leftrightarrow w \leftrightarrow u \leftrightarrow v$, is called a triangle.
2. **Rectangles:** Let b_0, b_1, c_0 , and c_1 be non-separating curves in Σ_g with $i(b_0, b_1) = i(c_0, c_1) = 1$ and all other pairs being disjoint. If $v_{ij} = (b_i, c_j, a_2, \dots, a_{g-1}) \in \mathcal{HT}^1(\Sigma_g)$ then the cycle, $v_{00} \leftrightarrow v_{01} \leftrightarrow v_{11} \leftrightarrow v_{10} \leftrightarrow v_{00}$, is called a rectangle.
3. **Pentagons:** Let $b_0, \dots, b_4$ be non-separating curves in Σ_g so that $i(b_i, b_{i+1}) = 1$, where the indices are modulo 5. The vertices $v_i = (b_i, b_{i+2}, a_2, \dots, a_{g-1})$ form a cycle, $v_0 \leftrightarrow v_2 \leftrightarrow v_4 \leftrightarrow v_1 \leftrightarrow v_3 \leftrightarrow v_0$, called a pentagon.

Attach a 2-cell along every cycle of type 1, 2, and 3 in $\mathcal{HT}^1(\Sigma_g)$ to construct the Hatcher–Thurston complex, $\mathcal{HT}(\Sigma_g)$.

Theorem 1.2 (Theorem 1.1, [HT80]). *$\mathcal{HT}(\Sigma_g)$ is connected and simply connected.*

Consider now, an infinite-type surface S with a non-planar end. There are two obvious issues with blindly generalizing the Hatcher–Thurston complex for S . Firstly, not all chains of cut systems have a maximal element when ordered by inclusion, for instance there exists a countable disjoint collection of non-separating curves in the Lochness monster such that the union of all the curves is separating but no finite subcollection is separating. Secondly, two cut systems with infinitely many distinct curves will not be connected by finitely many elementary moves. Thus we define the following complex by limiting cut systems to finitely many curves.

Definition 1.3. Let S be an infinite-type surface with non-planar ends, $k \in \mathbb{N}$, and let $\Gamma_k^0(S)$ be the collection of cut systems on S with exactly k curves. Let $\Gamma_k^1(S)$ be the graph obtained by attaching a 1-cell between two cut systems in $\Gamma_k^0(S)$ whenever they are related by an elementary move. The generalized Hatcher–Thurston complex, $\Gamma_k(S)$, is obtained by attaching 2-cells to every cycle of type 1, 2, or 3.

Note that $\Gamma_1^1(S)$ is just the Schmutz graph $\mathcal{G}(S)$. Since the Schmutz graph is connected and $\text{diam}(\mathcal{G}(S)) \leq 4$ we conclude that $\Gamma_1^1(S)$ is connected with finite diameter. This will be useful in the in the proof of connectedness of $\Gamma_k(S)$.

2 CONNECTEDNESS AND SIMPLE CONNECTEDNESS OF $\Gamma_k(S)$

In this section we prove Theorem 1.

Lemma 2.1. Let $l < k \in \mathbb{N}$ and v, w be cut systems in $\Gamma_k(S)$ so that v and w have all but one curve in common. Then there is a path of length at most 4 from v to w in $\Gamma_1^1(S)$.

Proof. Let $v - w = \{a\}$ and $w - v = \{b\}$ and let R be the surface obtained by cutting S along the representatives of all the common curves, $c_1, \dots, c_{k-1}$, of v and w . There is a path $a \leftrightarrow d_1 \leftrightarrow \dots \leftrightarrow d_n \leftrightarrow b$ in $\Gamma_1^1(R) = \mathcal{G}(R)$ with $n \leq 2$. Since d_i are non-separating curves in R it follows that $v_i = (d_i, c_1, \dots, c_{k-1})$ is a cut system. Thus, $v \leftrightarrow v_1 \leftrightarrow \dots \leftrightarrow v_n \leftrightarrow w$ is the desired path. $\square$

Proof of first half of Theorem 1. Let $v, w \in \Gamma_k^0(S)$. We use induction on the number of curves $\ell = |v - v \cap w|$. Lemma 2.1 shows that for the base case $\ell = 1$ there is a path from v to w . Suppose now that $v - w = \{a_0, \dots, a_{\ell-1}\}$, $w - v = \{b_0, \dots, b_{\ell-1}\}$, and $v \cap w = \{c_0, \dots, c_{k-\ell-1}\}$. Let $K \subset S$ be a compact sub-surface containing all the curves in v and w and let d be a non-separating curve in some component of $S - K$. Then, using induction, there is a path from $v' = (d, a_1, \dots, a_{\ell-1}, c_0, \dots, c_{k-\ell-1})$ to $w' = (d, b_1, \dots, b_{\ell-1}, c_0, \dots, c_{k-\ell-1})$. Using Lemma 2.1, there is a path from v to v' and w to w' . Using the construction above and the fact that the length

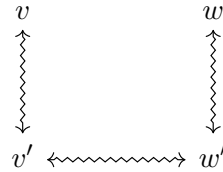


Figure 1: Constructing the path from v to w .

of the path in Lemma 2.1 is at most 4, it follows that the diameter of $\Gamma_k^1(S)$ is at most $8k - 4$. $\square$

If v, w are two cut systems in $\Gamma_k(S)$ so that all of the curves in $v \cup w$ are distinct then any path in $\Gamma_k^1(S)$ between v and w has length at least k . This means that the diameter of $\Gamma_k^1(S)$ has a lower bound k .

In order to prove that $\Gamma_k(S)$ is simply-connected we define the following notion of radius of a path, and segments.

Definition 2.2. The radius of a path $p = v_0 \leftrightarrow \dots \leftrightarrow v_n$ in $\Gamma_k^1(S)$ with respect to a non-separating curve, $a \in v_j$ for some j , is defined as $\max_{v_i} (\min_{b \in v_i} i(a, b))$. If $a_0, \dots, a_n$ are non-separating curves in S then p is a $a_0, \dots, a_n$ -segment if $a_0, \dots, a_n \in v_i$ for each vertex v_i in the path p .

First we show some propositions and lemmas in order to prove that $\Gamma_1(S)$ is simply-connected. The first one we will require is a special case of a well-known result about Dehn twists and intersection numbers.

Proposition 2.3 (Proposition 3.4, [FM11]). *Let a, b, c be arbitrary isotopy classes of curves in S and let T_a be the Dehn twist about a . Then for any $k \in \mathbb{Z}$,*

$$\left| i(T_a^k(b), c) - |k| i(a, b) i(a, c) \right| \leq i(b, c)$$

□

Note that vertices in $\Gamma_1(S)$ are just non-separating curves. A path is just a finite sequence, $a_0 \leftrightarrow a_1 \leftrightarrow \dots \leftrightarrow a_n$, of non-separating curves. We first show the following lemma.

Proposition 2.4 (Lemma 9, [Waj99]). *Let $p = a_0 \leftrightarrow a_1 \leftrightarrow a_2 \leftrightarrow a_3 \leftrightarrow a_0$ be a closed path in $\Gamma_1(S)$ so that $i(a_1, a_3) = 0$. Then p is null-homotopic.*

Proof. Note that for some $s \in \{1, -1\}$ the Dehn twist $T_{a_1}^s$ satisfies $i(T_{a_1}^s(a_2), a_0) < i(a_2, a_0)$, for any $k \in \mathbb{Z}$ the intersection $i(T_{a_1}^k(a_2), T_{a_1}^{k \pm 1}(a_2)) = 1$, and for any $k \in \mathbb{Z}$ the intersection $i(T_{a_1}^k(a_2), a_1) = i(T_{a_1}^k(a_2), a_3) = 1$, using the identity in Proposition 2.3. This means that for some $s \in \{1, -1\}$ and $k > 0$ define $p_k = a_0 \leftrightarrow a_1 \leftrightarrow T_{a_1}^{sk}(a_2) \leftrightarrow a_3 \leftrightarrow a_0$ to be a sequence of closed paths with $i(T_{a_1}^{sk}(a_2), a_0) < i(T_{a_1}^{sk-s}(a_2), a_0)$ and p_k and p_{k+1} are homotopic as $p_{k+1} - p_k$ can be split into two triangle paths. Let p_N be the path where the first and third curves are disjoint, then it follows by induction that $p = p_0$ and p_N are homotopic. Thus without loss of generality we now assume that $i(a_0, a_2) = 0$ in p . Now define $b = T_{a_1}(a_2)$. Using Proposition 2.3 it can be shown that $i(a_i, b) = 0$.

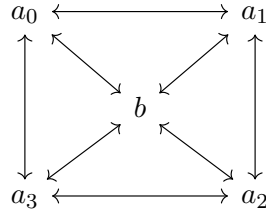


Figure 2: Splitting the closed path p into triangles.

Thus the path p can be split into four paths of type 1, see Figure 2. Hence p is null-homotopic. □

Lemma 2.5. *Every closed path $p = a_0 \leftrightarrow \dots \leftrightarrow a_{n-1} \leftrightarrow a_0$ in $\Gamma_1(S)$ of radius 1 with respect to a_0 is null-homotopic.*

Proof. Since p has radius 1 with respect to a_0 it follows that $i(a_0, a_i) \leq 1$ for all i . In the case that $i(a_0, a_i) = 1$ for all $i > 0$ it follows that the path p can be split into $n - 2$ paths of type 1, and hence conclude that p is null-homotopic. Now consider the case when $i(a_0, a_j) = 0$ for exactly one $j \neq 0$ and $i(a_0, a_i) = 1$ for $i \neq 0, j$. In this case p can be split into paths of type 1 and the closed path $q = a_0 \leftrightarrow a_{j-1} \leftrightarrow a_j \leftrightarrow a_{j+1} \leftrightarrow a_0$. Since paths of type are null-homotopic and q is null-homotopic by Proposition 2.4 it follows that p is null homotopic.

Finally, consider the case where the intersection number of a_0 with some a_i are 0 and with the rest it is 1. Then the path p can be split into paths of type 1, paths of the type in Proposition

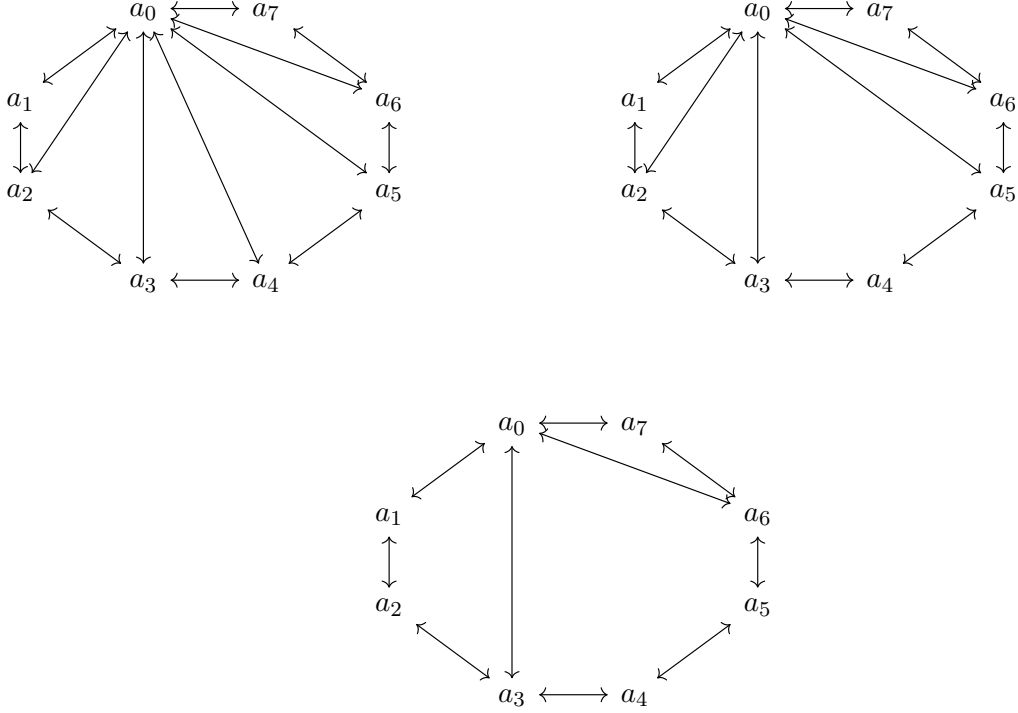


Figure 3: Examples of the three cases in proof of Lemma 2.5.

2.4, and some larger paths. Paths of the first two types are null homotopic, by definition of $\Gamma_1(S)$ and Proposition 2.4 respectively, thus we only have to show that paths of the third type are null-homotopic. Let $q = a_0 \leftrightarrow a_j \leftrightarrow a_{j+1} \leftrightarrow \dots \leftrightarrow a_{j+m} \leftrightarrow a_0$ be a path of the third type. Then we know that $i(a_0, a_{j+s}) = 0$ for all $0 \leq s \leq m$. Using the same argument as in Proposition 2.4 we can assume without loss of generality that $i(a_j, a_{j+2}) = 0$. Let $b = T_{a_j}(a_{j+1})$, then using the intersection identity in 2.3 it follows that $i(b, a_0) = i(b, a_{j+2}) = i(b, a_{j+1}) = 1$. This means that q is homotopic to the path $a_0 \leftrightarrow a_j \leftrightarrow b \leftrightarrow a_{j+2} \leftrightarrow \dots \leftrightarrow a_{j+m} \leftrightarrow a_0$ which, due to $a_0 \leftrightarrow a_j \leftrightarrow b \leftrightarrow a_0$ being a path of type 1, is homotopic to the path $q_1 = a_0 \leftrightarrow b \leftrightarrow a_{j+2} \leftrightarrow \dots \leftrightarrow a_{j+m} \leftrightarrow a_0$. Repeating this process we eventually arrive at a path q_n which is of the type in Proposition 2.4, which is null-homotopic. This completes the proof. $\square$

Proposition 2.6. *Every closed path $p = a_0 \leftrightarrow \dots \leftrightarrow a_{n-1} \leftrightarrow a_0$ in $\Gamma_1(S)$ is null-homotopic.*

Proof. The idea of the proof is to construct a curves $\beta_0, \beta_1, \beta_2$ in S so that $i(b_2, a_j) = 0$ for all j and $i(b_2, b_0) = i(b_2, b_1) = i(b_0, a_0) = i(b_1, a_1) = 1$. This means that the path $q = b_2 \leftrightarrow b_1 \leftrightarrow a_1 \leftrightarrow \dots \leftrightarrow a_{n-1} \leftrightarrow a_0 \leftrightarrow b_0 \leftrightarrow b_2$ has radius 1 with respect to b_2 . Since the path $b_2 \leftrightarrow b_1 \leftrightarrow a_1 \leftrightarrow a_0 \leftrightarrow b_0 \leftrightarrow b_2$ is also radius 1 with respect to b_2 it follows from Lemma 2.5 that q is homotopic to p and that q is null homotopic. It follows that p is null-homotopic.

Now we describe the construction of the curves β_i . Let R be a subsurface of S with compact boundary and exactly one planar end so that all the curves $\alpha_0, \dots, \alpha_{n-1}$ are contained in R . Let δ be a separating curve in R disjoint from all α_i so that the two surfaces, R_1 and R_2 , obtained by cutting R along δ satisfy the following: R_1 is compact and R_2 is of infinite-type with a compact boundary component corresponding to the curve δ in R and one non-planar end. Let τ_0 and τ_1 be homotopically distinct arcs in R_2 with their end points lying on the boundary of R_2 so that

$\tau_0 \cap \tau_1 = \emptyset$. Since δ and α_0 are disjoint and $i(a_0, a_1) = 1$, it can be shown that there exist disjoint essential arcs, σ_0, σ_1 , with end points on the boundary of R_1 corresponding to δ and $|\sigma_0 \cap \alpha_0| = 1$ and $|\sigma_1 \cap \alpha_1| = 1$. Gluing R_1 and R_2 together along the boundary corresponding to δ in such a way that $\beta_0 = \sigma_0 \cup \tau_0$ and $\beta_1 = \sigma_1 \cup \tau_1$ are closed curves. Then by construction we have $i(b_0, a_0) = i(b_1, a_1) = 1$ and that b_i are non-separating. Let β_2 be a non-separating curve in R_2 which intersects τ_0 and τ_1 exactly once. Thus we have our desired curves b_0, b_1 , and b_2 . $\square$

The above proposition implies that $\Gamma_1(S)$ is simply-connected when S has non-planar ends. We use this as our base case for induction on k , to show that $\Gamma_k(S)$ is also simply-connected. From here on we assume the induction hypothesis: for every surface S the complex $\Gamma_j(S)$ is simply connected for all $j < k$. We prove the simple connectedness now by also inducting on radius of the path. We start by a special case of radius 0 curves.

Proposition 2.7. *Suppose that $p = v_0 \leftrightarrow \dots \leftrightarrow v_{n-1}$ is a closed path in $\Gamma_k(S)$ so that all the vertices v_i contain a common curve c . Then p is null-homotopic.*

Proof. Let p' be the path in $\Gamma_{k-1}(S)$ where the vertices are $v_i - \{c\}$. By the induction hypothesis, it follows that p' is null-homotopic, meaning that it can be split into paths of type 1, 2, and 3. Since adding c back to each vertex in p' does not remove any edge connections therefore p is null-homotopic. $\square$

Proposition 2.8 (Lemma 13, [Waj99]). *Let a_0, a_1, a_2 be disjoint non-separating curves so that:*

1. *all unions $\alpha_i \cup \alpha_j$ are non-separating in S .*
2. *the union $\alpha_0 \cup \alpha_1 \cup \alpha_2$ separates S .*

Then there exists disjoint non-separating curves b_0, b_1, b_2 so that $i(a_i, b_i) = 1, i(a_{i+1}, b_i) = 1$ (indices modulo 3), and $i(a_i, b_{i+1}) = 0$ for all other i, j . Moreover, if $(a_0, a_1, c_2, \dots, c_{k-1})$ is a vertex in $\Gamma_k(S)$ then the closed path $p = v_0 \leftrightarrow \dots \leftrightarrow v_5 \leftrightarrow v_0$ defined as

$$\begin{aligned} (a_0, a_1, c_2, \dots, c_{k-1}) &\leftrightarrow (a_0, b_1, c_2, \dots, c_{k-1}) \leftrightarrow (a_0, a_2, c_2, \dots, c_{k-1}) \leftrightarrow (b_0, a_2, c_2, \dots, c_{k-1}) \\ &\leftrightarrow (a_1, a_2, c_2, \dots, c_{k-1}) \leftrightarrow (a_1, b_2, c_2, \dots, c_{k-1}) \leftrightarrow (a_0, a_1, c_2, \dots, c_{k-1}) \end{aligned}$$

is null-homotopic in $\Gamma_k(S)$.

Proof. Due to conditions on a_0, a_1, a_2 it follows that $S - \bigcup_i \alpha_i$ has exactly two connected components, R_1 and R_2 , both with at least three connected components corresponding to the curves α_0, α_1 , and α_2 . Let δ_i be disjoint arcs in R_1 with endpoints on the boundary components corresponding to α_i and α_{i+1} (indices taken modulo 3). Similarly define disjoint arcs δ'_i on R_2 . Defining $\beta_i = \delta_i \cup \delta'_i$ we get disjoint non-separating curves b_0, b_1, b_2 with the desired intersections with a_i . In order to prove the second part of the proposition, define $c = T_{a_1}(b_0)$. Then it follows from Proposition 2.3 that $i(c, b_0) = i(c, a_1) = i(c, b_1) = i(c, a_0) = 1$ and $i(c, b_2) = i(c, a_2) = 0$. If $(a_0, a_1, c_2, \dots, c_{k-1})$ is a vertex in $\Gamma_k(S)$ then define the vertices $w_0 = (d_1, d_2, c_2, \dots, c_{k-1}), w_1 = (c, d_2, c_2, \dots, c_{k-1}), w_2 = (c, a_2, c_2, \dots, c_{k-1})$. Then the path $p = v_0 \leftrightarrow \dots \leftrightarrow v_5 \leftrightarrow v_0$ as defined above can be split into paths of type 1, 2, and 3 as in Figure 4. $\square$

Before beginning the proof of the next lemma, note that if v and w are vertices in $\Gamma_k(S)$ with common curves $(a_0, \dots, a_m)$ then there is an $(a_0, \dots, a_m)$ -segment connecting v and w . This follows directly from the proof of connectedness. This fact will be repeatedly used in the proof of the next lemma.

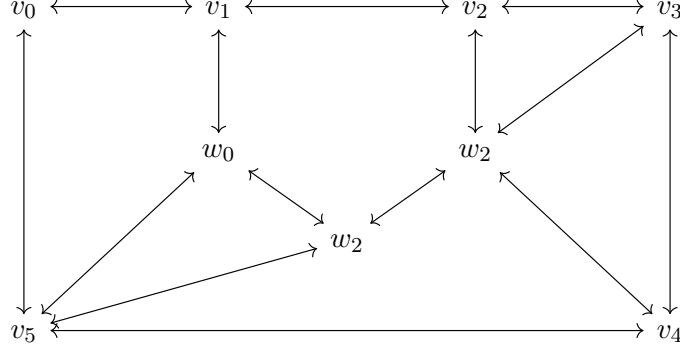


Figure 4: Splitting the path p into paths of type 1, 2, and 3.

Lemma 2.9. *Let p be a path in $\Gamma_k(S)$ of radius 0 with respect to some curve a_0 . Then p is null-homotopic.*

Proof. We prove this by induction on the number of segments in the path p . Suppose that $a_0 \in v_0$, where v_0 is a vertex in p . We call the maximal a_0 -segment containing v_0 the first segment. From Proposition 2.7 it follows that if p is a a_0 -segment then p is null homotopic. Suppose this is not the case and v_1 is the terminal vertex of the first segment. Let $a_1 \in v_1$ be a curve distinct from a_0 which will serve as the common curve of the next segment, which we call the second segment. Let v_2 be the terminal element of second segment. If p is just made up of these two segments then both v_1 and v_2 contain a_0 and a_1 , thus there is a (a_0, a_1) -segment from v_1 to v_2 , therefore implying that p can be split into a a_0 -segment and a a_1 -segment, both of which are null-homotopic due to Proposition 2.7. Suppose now that p has more than 2 segments. Then let a_2 be a curve in the vertex v which follows v_2 , so that $i(a_0, a_2) = 0$. Let v_3 be the terminal vertex in the a_2 -segment. Now we try to reduce the number of segments.

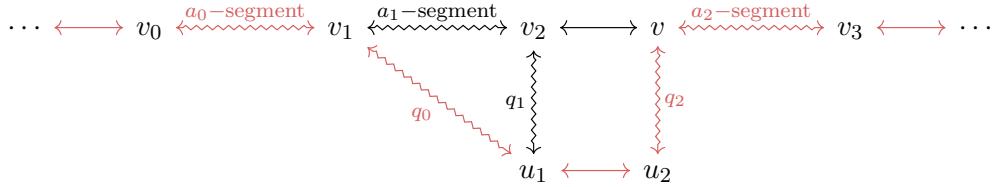


Figure 5: The construction in Case 1. Here q_0 is a (a_0, a_1) -segment, q_1 is a a_1 -segment, and q_2 is a a_2 -segment. The path in red has one less segment than p .

Case 1: Suppose that $a_2 \notin v_2$. Since $v_2 \leftrightarrow v$, it follows that $v - v_2 = \{a_2\}$ and $v_2 - v = \{a_1\}$, implying that $i(a_1, a_2) = 1$. Let $a_1 \cap a_2 = \{b_1, \dots, b_{k_1}\}$. This means that we obtain a connected surface, R , by cutting S along $a_1 \cup a_2$. Moreover the image of the curve a_0 in R is non-separating. Let $u = (b_1, \dots, b_{k_1})$ and u' be a vertex in $\Gamma_{k-1}(R)$ containing a_0 . By connectedness, it follows that there is a path q between u and u' . Let q_1, q_2 be paths in $\Gamma_k(S)$ obtained by adding a_1, a_2 respectively to each vertex of q . The path q_1 connects v_2 to u_1 and q_2 connects v to u_2 , where u_i contains both a_i and a_0 . Note that corresponding vertices v', w' in paths q_1, q_2 respectively are connected by an edge as $i(a_1, a_2) = 1$. Thus, observe that if $v' \leftrightarrow v''$ in q_1 and $w' \leftrightarrow w''$ in q_2 then the path $v' \leftrightarrow w' \leftrightarrow w'' \leftrightarrow v'' \leftrightarrow v'$ is a path of type 2. It follows from this that the closed

path which starts at v_2 moves along q_1 till u_1 , goes to u_2 via the connecting edge, $u_1 \leftrightarrow u_2$, then moves back up along q_2 to v and returns back to v_2 via the connecting edge, is null-homotopic. Also since v_1 and u_1 contain curves a_0, a_1 there is a (a_0, a_1) –segment, q_0 , connecting v_1 to u_1 . The closed path which starts from v_1 and moves along p till v_2 , then moves along q_1 to u_1 , and finally moves back up to v_1 along q_0 is also null-homotopic as it is a a_1 –segment. Now, the red path, p' , in Figure 5 is homotopic to p and bypasses the a_1 –segment in p , thus p' has one less segment than p .

Case 2: Suppose now that $a_2 \in v_2$ and $\alpha_0 \cup \alpha_2$ is non-separating. In the case when $\alpha_0 \cup \alpha_1 \cup \alpha_2$ is non-separating then there is a vertex $v' \in \Gamma_k(S)$ so that v' contains a_0, a_1, a_2 . Then there is a (a_0, a_1) –segment from v_1 to v' and a (a_1, a_2) –segment from v_2 and v' . These are shown in Figure 6. The red path in Figure 6 is homotopic to p as the closed path $v_1 \rightsquigarrow v' \rightsquigarrow v_2 \rightsquigarrow v_1$ is null-

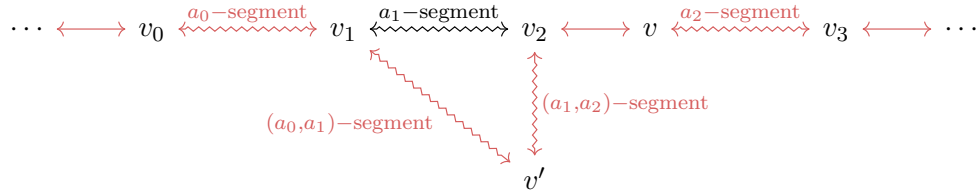


Figure 6: The red path is homotopic to p .

homotopic by Proposition 2.7. The red path has fewer segments than p as we have bypassed the second segment.

Now consider the case when $\alpha_0 \cup \alpha_1 \cup \alpha_2$ are separating. Let $(b_2, \dots, b_{k-1})$ be a cut system in the component of $S - (\alpha_0 \cup \alpha_1 \cup \alpha_2)$ which has a non-planar end. Then let $w_0 = (a_0, a_1, b_2, \dots, b_{k-1})$, $w_1 = (a_1, a_2, b_2, \dots, b_{k-1})$, and $w_2 = (a_2, a_0, b_1, \dots, b_{k-1})$ be vertices in $\Gamma_k(S)$. By Proposition 2.8, there is a null-homotopic closed path $q = w_0 \rightsquigarrow w_1 \rightsquigarrow w_2 \rightsquigarrow w_0$ where the path $w_i \rightsquigarrow w_{i+1}$ is a a_{i+1} –segment (indices modulo 3) for $i = 0, 1, 2$. Moreover, there is a (a_0, a_1) –segment, q_1 , from v_1 to w_0 and a (a_1, a_2) –segment, q_2 , from v_2 to w_1 . The closed path $v_1 \rightsquigarrow w_0 \rightsquigarrow w_1 \rightsquigarrow v_2 \rightsquigarrow v_1$ as shown in Figure 7 is also null-homotopic as it is a a_1 –segment, by Proposition 2.7. It follows that the red is homotopic to p and has fewer segments than p .

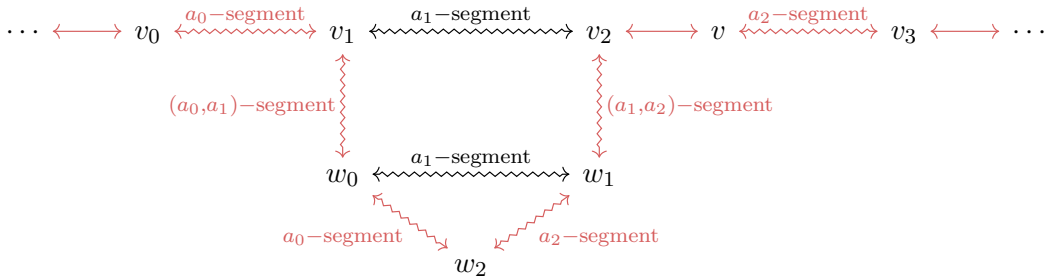


Figure 7: The red path is homotopic to p .

Case 3: Now suppose that $a_2 \in v_2$ and $\alpha_0 \cup \alpha_2$ separates S . Suppose that p has only three segments. The first vertex of the of first segment then contains both a_0 and a_2 as $i(a_0, a_2) = 0$. But this implies that $\alpha_0 \cup \alpha_2$ is non-separating, which is a contradiction. Thus p must contain at

least four segments in this case. We assume from here on that closed paths with radius 0 and up to three segments are null homotopic. Let a_3 be a non-separating curve in the vertex following v_3 so that $i(a_0, a_3) = 0$. Note that $i(a_2, a_3) \neq 1$ as α_2 is a separating curve in the surface obtained by cutting S along α_0 . Since v_3 is the last vertex of the third segment and a_3 cannot be involved in the move from v_3 to the next vertex, it follows that $a_3 \in v_3$. This also implies that $\alpha_2 \cup \alpha_3$ is non-separating. Now, let S_0, S_1 be the two surfaces obtained by cutting S along $\alpha_0 \cup \alpha_2$ and without loss of generality assume that α_4 is a curve in S_1 . Note that $\alpha_0 \cup \alpha_3$ is non-separating in S since otherwise $\alpha_2 \cup \alpha_3$ separates in S , a contradiction. Since $i(a_1, a_0) = i(a_1, a_2) = 0$ we can break this further down into two cases: either α_1 is contained in S_0 or α_1 is contained in S_1 .

Suppose that α_1 is in S_0 . Then $\alpha_0 \cup \alpha_1 \cup \alpha_3$ is non-separating in S as α_1 is non-separating in S_0 and α_3 is non-separating in S_1 . This means that there is a vertex $w = (a_0, a_1, a_3, b_3, \dots, b_{k-1})$ in $\Gamma_k(S)$. There segments from v_1, v_2, v_3 to w as shown in Figure 8. The red path depicted in Figure 8 is homotopic to p by Proposition 2.7 and has fewer segments than p .

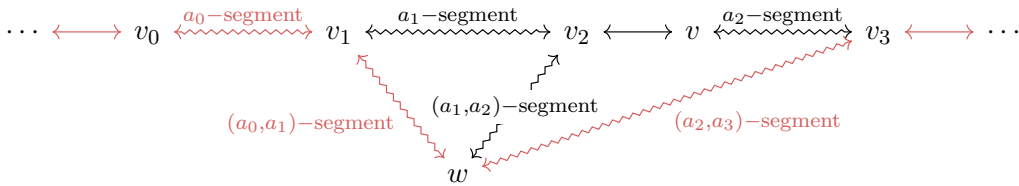


Figure 8: The red path is homotopic to p .

Now suppose that α_1 is contained in S_1 . If S_1 is a surface of finite-type with genus g then let w, w' be maximal cut systems in $\mathcal{HT}(S_1)$ containing a_1, a_3 respectively and let $t \in \Gamma_{k-g-1}(S_0)$ be some fixed vertex. If S_1 is infinite-type then let $w, w' \in \Gamma_{k-1}(S_1)$ be vertices containing a_1, a_3 respectively and let t be empty. In either case, from connectivity, it follows that there is a path q in $\Gamma_{k-1}(S)$ from w to w' . Let q_1 be the path in $\Gamma_k(S)$ obtained by adding a_0 and the curves in t to each vertex in q , and let q_2 be the curve obtained by adding a_3 and the curves in t to each vertex in q . The end points, w_2 and w'_2 , of q_1 contain the curves a_0, a_1 and a_0, a_3 respectively. Similarly there are end points, w_3 and w'_3 , of q_2 contain the curves a_2, a_1 and a_2, a_3 respectively. Figure 9 shows segments connecting v_i to w_i for $i = 2, 3$, v_1 to w'_2 , and w'_3 to v_4 . The closed paths $v_1 \rightsquigarrow v_2 \rightsquigarrow w_2 \rightsquigarrow w'_2 \rightsquigarrow v_1$, $v_2 \rightsquigarrow w_2 \rightsquigarrow w_3 \rightsquigarrow v_3 \rightsquigarrow v_2$, and $v_4 \rightsquigarrow v_3 \rightsquigarrow w_3 \rightsquigarrow w'_3 \rightsquigarrow v_4$ are null homotopic as their have zero radius and have at most three segments. Connecting corresponding edges in q_1 and q_2 by a path, we can split the closed path $w_2 \rightsquigarrow w_3 \rightsquigarrow w'_3 \rightsquigarrow w'_2 \rightsquigarrow w_2$ into closed paths with a_0 -radius 0 and consisting only 3 segments, therefore concluding that this closed path is also null-homotopic. Thus the red path depicted in Figure 9 is homotopic to p and has fewer segments. $\square$

Now we are finally ready to prove Theorem 1.

Proof of Theorem 1. Suppose that $p = v_0 \leftrightarrow \dots \leftrightarrow v_{n_1} \leftrightarrow v_0$ is a path in $\Gamma_k(S)$. Let K be a compact set containing all the curves in the vertices of p and let b be a non-separating curve in $S - K$. Let $a_0 \in v_0$ and $a_1 \in v_1$ be disjoint curves and let w_0 and w_1 be vertices in $\Gamma_k(S)$ which contain b, a_0 and b, a_1 respectively. The vertices w_0 and v_0 can be connected by a a_0 -segment, w_1 and v_1 can be connected by a a_1 -segment, and w_0 and w_1 can be connected by a b -segment. The closed path $v_0 \leftrightarrow w_0 \leftrightarrow w_1 \leftrightarrow v_1 \leftrightarrow v_0$ has a_0 -radius 0 implying that the red path in Figure 10 is

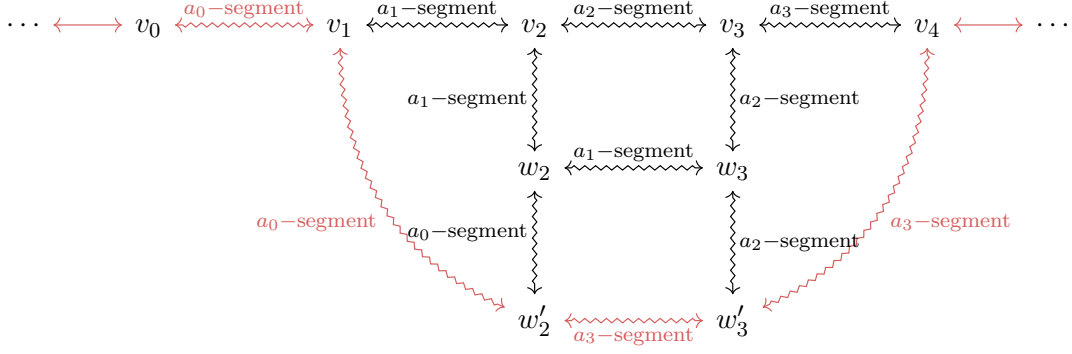


Figure 9: The red path is homotopic to p .

homotopic to p . Since the red path has b -radius 0 it follows that it is null-homotopic and thus p is null-homotopic. $\square$

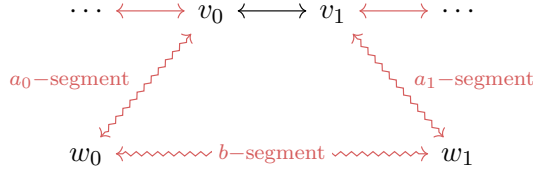


Figure 10: The red path is homotopic to p .

3 AUTOMORPHISMS OF $\Gamma_k(S)$ AND RIGIDITY

In this section we prove Theorem 2 and henceforth assume that S has no planar ends and boundary. We state the results proved in [IK07] without proof whenever the proof can be directly translated to the infinite-type case without changes. Using Theorem 1.1 we already have that $\text{Aut}(\Gamma_1(S)) \cong \text{Map}^*(S)$, as $\Gamma_1^1(S) = \mathcal{G}(S)$. First we describe an action of $\text{Aut}(\Gamma_k(S))$ for $k \geq 2$ on the Schmutz graph, $\mathcal{G}(S)$.

Action of $\text{Aut}(\Gamma_k(S))$ on $\mathcal{G}(S)$. Let a be a non-separating curve and $f \in \text{Aut}(\Gamma_k(S))$. Let $a_1, \dots, a_{k-1}$ be curves in S so that $v = (a, a_1, \dots, a_{k-1})$ is a cut-system. Let b be another non-separating curve with $i(a, b) = 1$ so that $w = (b, a_1, \dots, a_{k-1})$ is also a cut-system. By construction, $v \leftrightarrow w$ and since f is simplicial it follows that $f(v) \leftrightarrow f(w)$. Define $\tilde{f}(a)$ as the unique curve in $f(v) - f(w)$. The following lemma states that $\tilde{f}$ is well defined.

Lemma 3.1 (Lemma 5, [IK07]). *The curve $\tilde{f}(a)$ is independent of all choices.* $\square$

The proof of the above lemma involves several steps. First, for a given non-separating curve a the subcomplex, $X_a \subset \mathcal{G}(S)$, whose vertices are curves which intersect a exactly once is shown to be connected following the proof of Lemma 3 in [IK07]. This fact along with the first half of Theorem 1 is used to show that $\tilde{f}(a)$ is independent of the choices of b and $a_1, \dots, a_{k-1}$. Further, it is shown that $\tilde{f}$ is an automorphism of $\mathcal{G}(S)$.

Proposition 3.2 (Proposition 8, [IK07]). *The map $\tilde{f} : \mathcal{G}(S) \rightarrow \mathcal{G}(S)$ is an automorphism.*

Proof. Let a, b be non-separating curves so that $i(a, b) = 1$. There exists a collection of curves $a_1, \dots, a_{k-1}$ which are disjoint from both a and b such that $v = (a, a_1, \dots, a_{k-1})$ and $w = (b, a_1, \dots, a_{k-1})$ are cut systems. Thus $f(v) \leftrightarrow f(w)$ as $f \in \text{Aut}(\Gamma_k(S))$ and $v \leftrightarrow w$. Since by definition $\tilde{f}(a) = f(v) - f(w)$ and $\tilde{f}(b) = f(w) - f(v)$ it follows that $i(\tilde{f}(a), \tilde{f}(b)) = 1$. This shows that $\tilde{f}$ is a simplicial map. The inverse map to $\tilde{f}$ is given by $\tilde{f}^{-1}$, see Lemma 7 in [IK07]. $\square$

Next, we generalize Proposition 9 in [IK07]. Since $\tilde{f}$ can also be viewed as a map on the non-separating complex $\mathcal{N}(S)$, it is natural to ask whether it is an automorphism of $\mathcal{N}(S)$? The next proposition provides a positive answer.

Proposition 3.3. *The map $\tilde{f} : \mathcal{N}(S) \rightarrow \mathcal{N}(S)$ is also an automorphism.*

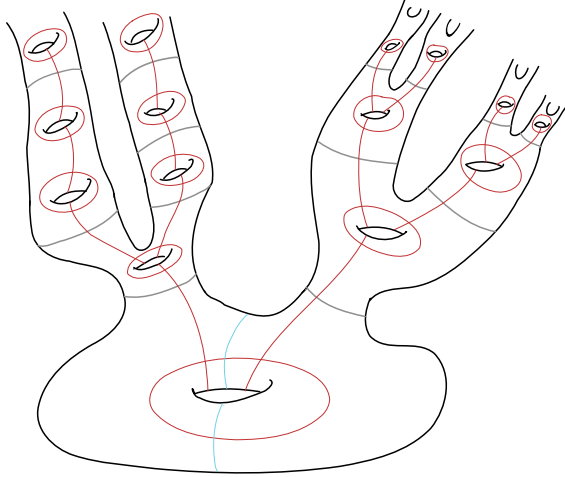
Proof. Using the same argument as before $\tilde{f}$ is a bijection. It only remains to show that it is also simplicial. Let $a, b \in \mathcal{N}(S)$ be two vertices so that $a \leftrightarrow b$, or in other words $i(a, b) = 0$. Consider the case when $\alpha \cup \beta$ is non-separating in S . Let $a_2, \dots, a_{k-1}$ be non-separating curves so that $(a, b, a_2, \dots, a_{k-1})$ is a vertex in $\Gamma_k(S)$. Since $\tilde{f}(a)$ and $\tilde{f}(b)$ are in $f(v)$ by definition it follows that $i(\tilde{f}(a), \tilde{f}(b)) = 0$.

Now suppose that now that $\alpha \cup \beta$ is separating in S . Then without loss of generality we can assume that they are as shown in the Figure 11a. We construct a system of curves on S in the following way:

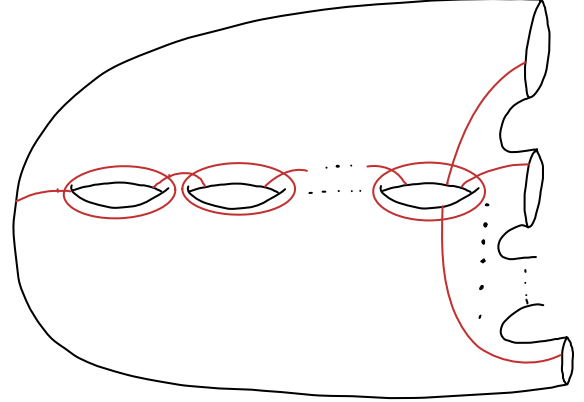
1. Let $\{K_n\}_{n \in \mathbb{N}}$ be an exhaustion of S by compact subsurfaces with K_1 being a genus 1 subsurface with two boundary components containing the curves α and β . It is always possible to construct an exhaustion so that all components of ∂K_n correspond to separating curves in S . Let C_1 be the system of curves and arcs in K_1 as shown in Figure 11b. Let γ_0 be the curve which intersects α and β exactly once.
2. Let $\Sigma_{g,b}$ be the topological type of a component, X , of $K_n - K_{n-1}$. Let C_X be a chain of filling non-separating curves and essential arcs in X , as shown in the Figure 11b. Let $C_n = \bigcup_X C_X$ be a system of curves on $K_n - K_{n-1}$ for $n \geq 2$.
3. Let C be a chain of curves in S consisting of all the curves in C_n for every n and the curves obtained by joining the two arcs which meet at every component of $\bigcup_n \partial K_n$. See Figure 11a for an example.

For any two curves $\gamma, \gamma' \in C$ either $i(c, c') = 0$ or $i(c, c') = 1$. Since $\gamma \cup \gamma'$ is non-separating in S , the non-separating case implies that $i(\tilde{f}(c), \tilde{f}(c')) = 0$ if $i(c, c') = 0$ and it follows from Proposition 3.2 that $i(\tilde{f}(c), \tilde{f}(c')) = 1$ if $i(c, c') = 1$. Similarly, when $\gamma \in C$ is such that $i(a, c) = i(b, c) = 0$ it follows that $i(\tilde{f}(a), \tilde{f}(c)) = i(\tilde{f}(b), \tilde{f}(c)) = 0$ and $i(\tilde{f}(a), \tilde{f}(c_0)) = i(\tilde{f}(b), \tilde{f}(c_0)) = 1$. Let $d = \tilde{f}(c)$ for each $\gamma \in C$ and let $d_0 = \tilde{f}(c_0)$. Then, as discussed above, the collection $\tilde{f}(C)$ forms a chain of non-separating curves in S . Note that $\tilde{f}(C)$ cannot lie inside any compact subsurface of S due the chain being infinitely large. Cutting S along $\tilde{f}(C) - d_0$ we get disjoint copies of an annuli A with a puncture in each boundary. Since $\tilde{f}(a), \tilde{f}(b)$ are disjoint from every $d \neq d_0$ it follows that they correspond to curves in some copy of the annuli A . Since $\tilde{f}(a) \neq \tilde{f}(b)$ it follows that they lie in distinct copies of A and thus are disjoint in S . $\square$

Proof of Theorem 2. Let $\Phi : \text{Aut}(\Gamma_k(S)) \rightarrow \text{Aut}(\mathcal{G}(S))$ be the map $f \mapsto \tilde{f}$. Suppose that $f, g \in \text{Aut}(\Gamma_k(S))$ then by Lemma 7 in [IK07] it follows that $\Phi(f \circ g) = \Phi(f) \circ \Phi(g)$. Suppose that $f \in \text{Ker}(\Phi)$ and let $v = (a_0, \dots, a_{k-1})$ be a vertex in $\Gamma_k(S)$ then by construction $\tilde{f}(a_i) \in f(v)$



(a) This figure shows an example of the construction of C on a surface with non-planar end space being the Cantor set union two isolated points. The blue curves are α and β , the gray curves are boundaries of ∂K_n , and the red curves are elements of C .



(b) The choice of chain on a surface of genus g and b boundary components.

Figure 11: Example of constructions of C_X and C .

for all i , and thus $f(v) = (\tilde{f}(a_0), \dots, \tilde{f}(a_{k-1})) = (a_0, \dots, a_{k-1}) = v$. This proves that Φ is a monomorphism. Moreover, Φ is surjective since if $h \in \text{Aut}(\mathcal{G}(S))$, then by Theorem 1.1 it follows that h is induced from a homeomorphism $F : S \rightarrow S$. Let $f = F_*$ be the induced automorphism on $\Gamma_k(S)$. If a is a non-separating curve, $v = (a, a_1, \dots, a_{k-1})$, and $w = (b, a_1, \dots, a_{k-1})$ are vertices in $\Gamma_k(S)$ as defined earlier then the unique curve in $f(v) - f(w)$ is just given by $F(a) = h(a)$ as f, h are induced by F . Thus Φ is an epimorphism. Hence we have $\text{Aut}(\Gamma_k(S)) \cong \text{Aut}(\mathcal{G}(S)) \cong \text{Map}^*(S)$. This completes the proof of Theorem 2. $\square$

Note that the map $\text{Map}^*(S) \rightarrow \text{Aut}(\Gamma_k(S))$ induced by the natural action of $\text{Map}^*(S)$ on $\Gamma_k(S)$ factors through the natural map $\text{Map}^*(S) \rightarrow \text{Aut}(\mathcal{G}(S))$ via Φ^{-1} . In particular, the map $\text{Map}^*(S) \rightarrow \text{Aut}(\Gamma_k(S))$ induced by the natural action is an isomorphism. Call this isomorphism $\Psi : \text{Map}^*(S) \rightarrow \text{Aut}(\Gamma_k(S))$.

Proof of Corollary 3. Further, if S and S' are infinite-type surfaces and let $h : \Gamma_k(S) \rightarrow \Gamma_k(S')$ be an isomorphism. Then the map, $h_* : \text{Aut}(\Gamma_k(S)) \rightarrow \text{Aut}(\Gamma_k(S'))$ induced by conjugation, i.e. $f \mapsto h \circ f \circ h^{-1}$, is an isomorphism. The composition $\Psi^{-1} \circ h_* \circ \Psi : \text{Map}^*(S) \rightarrow \text{Map}^*(S')$ is an isomorphism which, by Theorem 1.1 of [BDR20], is induced by composition by a homeomorphism $G : S \rightarrow S'$, i.e. $\Psi^{-1} \circ h_* \circ \Psi : [F] \mapsto [G \circ F \circ G^{-1}]$. Thus we have $h_*(F_*) = G_* \circ F_* \circ G_*^{-1}$, for every mapping class $[F]$. Since by theorem 2 every automorphism of $\text{Aut}(\Gamma_k(S))$ is induced by a mapping class we conclude that $h_*(f) = G_* \circ F_* \circ G_*^{-1}$ where $f = F_*$. This completes the proof of Corollary 3. $\square$

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